AI in Sustainability & Green Initiatives

Problem statement: A large Indian conglomerate wants to reduce its carbon footprint, optimise energy consumption, and meet sustainability goals. Propose an AI solution for energy optimisation, waste reduction, and environmental impact monitoring across its operations.

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❖ Problem statement Abstraction: -

- A large Indian conglomerate operates across multiple industries, such as manufacturing, energy, transportation, infrastructure, retail, and logistics. These operations consume significant amounts of electricity, fuel, water, and raw materials. They also generate industrial waste, greenhouse gas emissions, and other forms of environmental pollution.
- As the business expands, its energy requirements and environmental impact may increase. Traditional methods of monitoring energy consumption, managing waste, and measuring carbon emissions often rely on manual inspections, historical reports, and separate monitoring systems. These methods make it difficult to identify inefficiencies in real time and take corrective action quickly.
- Therefore, the conglomerate needs an AI-powered sustainability management system that uses real-time operational data, machine learning, predictive analytics, and intelligent automation to reduce its carbon footprint, optimize resource utilization, minimize waste, and achieve its long-term environmental sustainability goals.

❖ Major problems: -

1. High Energy Consumption and Carbon Emissions
   - Industrial machinery, manufacturing plants, warehouses, office buildings, and transportation networks consume substantial energy. Inefficient equipment, unnecessary machine operation, poor energy scheduling, and excessive cooling requirements can increase electricity consumption and operational costs. If this energy comes from fossil fuels, it also contributes to greenhouse gas emissions.
2. Inefficient Waste Management
   - Manufacturing processes can generate defective products, excess raw materials, packaging waste, and industrial by-products. Without accurate demand forecasting and waste tracking, organizations may overproduce goods, discard reusable materials, and send recyclable waste to landfills. This increases disposal costs and resource consumption.
3. Lack of Real-time Environmental Monitoring
   - Monitoring air pollution, carbon emissions, water consumption, wastewater quality, and industrial discharge across multiple locations is challenging. Periodic manual inspections may fail to identify abnormal conditions quickly, making it difficult to prevent environmental damage and maintain compliance with applicable regulations.

4. Limited Predictive and Data-Driven Decision-Making
   - Different départements may maintain separate datasets for energy, production, transportation, and waste management. Without integrated analytics, management may struggle to understand which facilities produce the most emissions, which machines waste the most energy, and which operational changes would provide the greatest environmental benefit.

❖ Proposed AI-Based Sustainability Solution

- To address sustainability challenges in large Indian conglomerates, an integrated sustainability management system is proposed to collect and analyze operational data from manufacturing plants, energy facilities, warehouses, and transportation networks. By combining IoT sensors, predictive analytics, and real-time monitoring, the system can identify energy inefficiencies, reduce waste, monitor environmental conditions, and track carbon emissions.
- The solution consists of three key components: energy optimization and carbon reduction, intelligent waste management, and real-time environmental impact monitoring. These components will operate through a central dashboard to support informed decision-making, improve resource utilization, reduce costs, and support long-term sustainability goals.

❖ Basic requirements for this solution

A. IoT Sensors and Smart Energy Meters: -
Purpose: - To collect real-time energy consumption and equipment-performance data.
- These devices provide the operational data required to identify energy wastage and understand equipment behavior.

B. Centralized Data Collections and Storage System: -
Purpose: - To store and integrate energy data from multiple facilities in one location.
- The organization needs a centralized data platform to collect information from sensors, production systems, utility bills, equipment records, and renewable-energy installations. The platform should maintain historical and real-time data in a consistent format.

C. AI and Machine Learning Models: -
Purpose: - To predict energy demand, detect inefficiencies, and recommend energy-saving actions.
- The system requires machine learning algorithms that analyze historical and real-time data. Time-series forecasting can predict future energy requirements, anomaly detection can identify unusual consumption, and optimization algorithms can recommend efficient equipment settings and operating schedules.

D. Integration with Industrial Equipment and Control Systems: -
Purpose: - To convert AI predictions into practical operational improvements.
- The organization needs compatible industrial control systems, such as PLCs, SCADA, or management systems, to implement approved recommendations. These systems can support adjustments to machine schedules, cooling settings, pumps, and other energy-consuming equipment.

E. AI Governance, Domain Experts and Human Oversight: -
Purpose: - To ensure that AI-based energy optimization decisions are accurate, safe, transparent, environmentally responsible, and approved by qualified domain experts.
- The organization needs an AI governance framework that defines rules for data quality, model validation, access control, cybersecurity, accountability, and continuous performance monitoring. AI models should be tested before deployment and periodically reviewed to ensure that their predictions remain reliable under changing operational conditions.
- Domain experts must work alongside AI systems. These experts will validate AI recommendations, assess operational risks, ensure compliance with environmental requirements, and approve changes that could affect machinery, production, or worker safety.

❖ AI for energy Optimization and Carbon Reduction

1. Objective: -The main objective is to reduce energy consumption and carbon emissions across manufacturing plants, warehouses, office buildings, and transportation networks by using AI-based prediction, monitoring, and optimization techniques.

2. Real-World Example: -

❖ ITC Limited's energy-efficiency initiatives:

- ITC's FY 2024-25 Sustainability Report documents several relevant measures: -
  1. Approximately 40TJ energy saving from energy-conservation equipment, equivalent to more than 5,000 tonnes of CO_2e emissions.
  2. Renewable energy accounted for 52% of total energy consumption.
  3. Renewable electricity assets had a reported capacity of 174 MWp.
  4. At its Bhadrachalam paper mill, a high-pressure recovery boiler reduced annual coal consumption by approximately 153,000 tonnes and avoided around 240,000 tonnes of CO_2e emission.

- Proposed solution AI-based Energy Management System: -

  The system will combine four main capabilities: energy demand forecasting, energy-wastage detection, intelligent equipment optimization, and carbon-emission monitoring. These capabilities will help the organization make data-driven decisions, improve equipment efficiency, reduce unnecessary energy consumption, and measure progress towards its sustainability objectives.

  How AI will be used for energy optimization

  A. Real-Time Energy Data Collection

  - The first step is collecting accurate operational data from different facilities. Smart meters and IoT sensors can measure electricity consumption, voltage, current, temperature, pressure, machine operating hours, fuel consumption, and production output.
  - For example, sensors installed on industrial motors and compressors can continuously measure their electricity consumption and operating conditions. The system can transmit this information to a central platform, allowing energy managers to monitor consumption across different facilities.

  B. AI-Based Energy Demand Forecasting Using a Hybrid Machine Learning Model

  - Proposed Solution
    - A single machine learning algorithm may not capture every pattern in industrial energy consumption because demand depends on historical usage, production volume, weather conditions, machine operating hours, shift schedules, equipment efficiency, and industry-specific processes.

  - Why Is a Hybrid Machine Learning Model Required?
    - Historical energy consumption: Previous electricity and fuel usage reveal daily, weekly, seasonal, and long-term consumption patterns.
    - Production volume: Higher production may require more electricity, fuel, cooling, and machinery operation.
    - Weather conditions: Temperature and humidity influence air conditioning, refrigeration, cooling towers, and other utilities.
    - Machine operating conditions: Equipment age, operating speed, maintenance condition, load, and efficiency affect energy consumption.

  - Expected Benefits
    - Improved forecasting of electricity and fuel requirements.
    - Better planning of industrial production and equipment schedules.
    - Earlier identifications of unexpected demand increase.
    - Improved coordination between energy demand and renewable-energy generation.
    - Reduced peak demand and electricity costs where operational conditions permit.
    - Better data-driven planning for energy efficiency and carbon reduction.

- AI Based Waste Reduction and Resource Optimization

  - Objective: The primary objectives are to develop an AI-powered Waste Reduction and Resource Optimization System that minimizes waste generation, improves raw-material utilization, reduces defective products, and increase recycling and reuse across conglomerate's manufacturing plants, warehouses, retail outlets, and logistics operations.

❖ Proposed Solution

- The organization should implement an integrated AI-based waste management system that collects production, Inventory, product-quality, raw-material consumption, and waste-generation data from different facilities.
- Machine learning models will analyze historical and real-time data to predict waste generation and identify inefficient processes. Computer vision will detect defective products and classify recyclable materials while optimization algorithms will improve material allocation, production planning, and recycling decisions.

❖ Five- Step Implementation Framework

- The system follows a clear sequence: Measure → Prevent → Recover → Convert → Verify. Prevention is given priority because material that is never wasted does not need recycling.

❖ Solution: Five steps, in this order

Step-1: Measure

- All major waste streams are recorded in tonnes and converted into CO_2e. The figures are divided by production volume so that business growth is not mistaken for failure. Machines, material lots and processes that create the highest waste and cost are identified first.

Step-2: Prevent

- Machine learning models study past production data and forecast scraps and defects. Computer vision systems detect defective products at an early stage. optimization tools improve cutting patterns and production plans while keeping quality and safety limits. Any change in machine settings is first tested in simulation and approved by engineers

- real-world evidence: Google's DeepMind reduced data center cooling energy use by up to 40%. In 2024, Google also reduced data center emissions by 12% despite a 27% increase in electricity demand, demonstrating the potential of AI-driven efficiency improvements.

Step-3: Recover

- For waste that cannot be avoided, camera-based systems sort materials and detect contamination. Collection routes are planned to reduce transport. Partnerships with recyclers are established before recovery claims are made.

Step-4 Convert

- Waste from one process is matched as raw material for another process. Matching considers material quality, distance and net environmental benefit. Long-distance transport that cancels the environmental gain is avoided.

Step-5 Verify

- Every claimed saving is compared with a similar production line that was not changed. Only material that has left the site for confirmed use is counted as recycled or diverted. The electricity used by the AI platform is included in the final calculation

- Controlling mistakes: Type I and Type II errors

  - Type I errors (false alarms) occur when the system incorrectly flags a normal situation, such as stopping a production line, penalizing a supplier, or reporting savings that did not actually occur. Since these actions can create significant operational and financial costs, false alarms must be carefully controlled. The appropriate level depends on the cost of the error. For alerts reviewed by people, higher sensitivity is acceptable because verification costs are relatively low. However, for automatic actions and reported sustainability claims, stricter thresholds should be applied to prevent unnecessary disruptions and inaccurate reporting.

  - Type II errors (missed detections) occur when the system fails to identify a real problem, such as equipment faults, hazardous waste, or environmental risks. Since these errors can have serious safety and compliance consequences, detection settings should remain sufficiently sensitive in high-risk areas.

- Pros, Cons and Loopholes

  - Pros

    - The solution addresses real problems faced by large Indian companies – high energy use, waste and lack of real-time data.
    - It combines IoT, machine learning and human oversight in a practical way.
    - Real examples from ITC and Reliance make the proposal relevant to the Indian context.
    - Phased implementation and verification reduce the risk of wrong claims.
    - Clear focus on prevention before recycling is environmentally sound.

  - Cons

    - Installing sensors and connecting old machines across many plants will take time and money.
    - Different factories have different energy and waste patterns, so one model cannot be used everywhere.
    - Employees may resist new recommendations if they are not involved in the process.
    - The AI system itself consumes electricity, which must be subtracted from claimed savings.

  - Possible Loopholes

    - If data from sensors is incomplete or wrong, the forecasts will also be wrong.
    - If the company claims savings without proper before-and-after measurement, the results can be questioned by auditors or regulators.
    - Over-dependence on automatic decisions without expert checking can create safety or quality problems.
    - Focusing only on waste reduction while ignoring energy or transport impact can shift the problem instead of solving it.

- Challenges

  1. Poor Data Quality and Missing Data

     - Challenge: Energy data collected from smart meters, IoT sensors, and industrial equipment may contain missing values, incorrect readings, duplicate records, or inconsistent timestamps.

- Example: A factory's electricity meter stops transmitting data for two hours. The model receives incomplete consumption data and may produce an inaccurate forecast.
- Solution: Apply data validation, missing-value handling, sensor-error detection, and timestamp synchronization before training the model.

2. Different Energy Patterns Across Industries

- Challenge: Different business units have different energy consumption patterns, so a model trained for one facility may not perform well at another.
- Example: A paper mill has continuous production and high steam demand, whereas a retail warehouse has energy demand influenced by refrigeration, lighting, and operating hours.
- Solution: Develop facility-specific models or industry-specific model configurations and validate them separately before deployment.

3. Difficulty Combining Multiple Models

- Challenge: A hybrid model does not automatically produce better forecasts. Different models may gen rate conflicting predictions, and an unsuitable combination can increase forecasting errors.
- Example: XGBoost predicts 12,000 kWh for the next operating day, while LSTM predicts 13,500 kWh and SARIMAX predicts 11,800 kWh.
- Solution: Combine predictions using validation-based weights or a trained meta-model. Compare the hybrid model against individual models using chronological test data and metrics such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).

4. Increasing Complexity of Industrial Waste

- Challenges: As industrial production expands and product designs become more complex, waste streams may contain a wider variety of materials, chemical compositions, composites and contaminated components. Conventional sorting systems and AI models trained on existing waste categories may struggle to classify these new materials accurately.
- Example: An automotive manufacturing facility may generate mixed plastic components, metal scrap, electronic waste and composite materials. A vision model trained in common metal and plastic categories may fail to identify newly introduced materials.
- Proposed solution: Develop adaptable computer vision models and material-classification systems using updated datasets. Combine image analysis with relevant sensor measurements and material records. Require specialist verification when material composition or hazardous properties cannot be reliably identified.

5. Environmental Trade-Offs and the Resource Footprint of AI

- Challenge: AI infrastructure itself consumes electricity, computing resources and equipment. In addition, reducing waste in one part of a process may increase energy consumption, water use or waste elsewhere. Focusing on one indicator may therefore shift rather than eliminate environmental impact.
- Example: An intensive material-sorting process may recover more recyclable material but consume enough additional energy or transport resources to reduce its overall environmental benefit.
- Proposed solution: Evaluate waste prevention and recovery alongside energy consumption, water use, transport and relevant emissions. Prefer prevention when feasible and compare recovery alternatives using lifecycle or other suitable environmental assessment methods. Include the AI system's own energy costs where material to the assessment.

- ❖ Expected impact: Better decisions based on total environmental performance rather than a single waste-recovery metric.

6. Changing Production Processes and Model Drift

   - ❖ Challenge: AI models trained on historical data may become less reliable when machinery, raw materials, suppliers, product designs or production schedules change. This problem, known as model drift, may cause the system to miss new waste patterns or generate unnecessary alerts.
   - ❖ Example: A defect-detection model trained on one product design may incorrectly classify components after the manufacturer introduces a redesigned product.
   - ❖ Proposed solution: Monitor model performance continuously, track changes in input data and retrain models when justified. Use representative validation datasets after production changes and require approval before deploying updated models. Maintain a fallback process when model reliability falls below an approved threshold.
   - ❖ Expected impact: Greater long-term reliability and reduced dependence on outdated predictions.

7. Balancing Waste Prevention, Production Quality and Cost

   - ❖ Challenge: Waste reduction cannot be optimized independently of production quality, equipment safety, delivery schedules and manufacturing costs. An AI system that minimizes material consumption without considering these constraints could increase defective products or disrupt production.
   - ❖ Example: A cutting-pattern optimization system may reduce material offcuts but produce components that fail dimensional or quality requirements if constraints are incorrectly specified.
   - ❖ Proposed solution: Use constrained optimization with explicit quality, safety, production and material specifications. Evaluate proposed changes in simulation or controlled trials before implementation. Reinforcement learning may be considered for complex operating decisions only after suitable simulation, safety constraints and engineering validation are established.
   - ❖ Expected impact: Lower avoidable waste while maintaining product quality and operational stability.

❖ Challenges overcome solution:

1. Establish a Reliable Data and AI Accuracy Management Strategy

   - Challenges addressed: Poor data quality, missing data, inaccurate predictions, and conflicting results from hybrid models.
   - Implementation strategy:
     - ➢ Apply automated data validation to detect missing, duplicate, or incorrect records before analysis.
     - ➢ Use facility-specific model configurations to account for differences in energy consumption patterns across industries.
     - ➢ Compare and validate XGBoost, LSTM, and SARIMAX models, combining predictions only when accuracy improves.
     - ➢ Continuously monitor performance using metrics such as MAE and RMSE against predefined thresholds.
     - ➢ Use confidence-based decision rules, with low-confidence or conflicting predictions reviewed by domain experts before action is taken.

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2. Implement Human-in-the-loop AI and a Continuous Feedback strategy

- Challenges addressed: Negative feedback from employees, rejection of AI recommendations, operational mistakes, and a lack of trust in automated decisions.
- Implementation Strategy:
  - Feedback collection: Provide options for employees to accept, reject, or modify AI recommendations, with a recorded reason for important rejections.
  - Feedback verification: Compare employee feedback with sensor readings, maintenance records, production data, and actual environmental outcomes.
  - Continuous improvement: Review repeated rejection patterns and update the system when evidence confirms that recommendations are impractical or incorrect

3. Develop an Integrated AI Governance, Knowledge and Security Strategy

- Challenges addressed: Fragmented information, difficulty identifying root causes, cybersecurity risks, unauthorized actions, and poor coordination between departments and facilities.
- Implementation strategy:
  - Centralized sustainability platform: Integrate energy, production, inventory, waste, transportation, and environmental data into a common platform with standardized definitions.
  - Knowledge graph: Link machines, raw materials, production batches, waste streams, emissions, suppliers, and recovery facilities to improve traceability and root-cause analysis.
  - Search-based diagnosis: Apply BFS to identify related factors and DFS to investigate deeper pathways, supported by causal analysis and expert validation before confirming causes.
  - Agentic AI coordination: Use AI agents to collect information, select approved analytical tools, compare recommendations, create action plans, and monitor outcomes, with risk-based permissions and human approval for significant actions.
  - Governance and accountability: Define ownership for data quality, model performance, safety, environmental reporting, and incident response, while maintaining records of recommendations, approvals, overrides, and verified outcomes.

❖ Financial Impact and Return on Investment (ROI)

- A large-scale AI sustainability system cannot be judged only by small investment figures. The correct approach is to start with a limited pilot in 2–3 plants.
  In the pilot phase, the company should measure the actual energy and waste costs of those plants before and after implementing the system. Only after independent verification of real savings should the system be expanded to other units.
- Long-term benefits will come from lower electricity bills, reduced waste disposal costs, and better material utilization. These benefits must be higher than the total cost of sensors, software, integration, training and maintenance. The electricity used by the AI system itself should also be included while calculating net savings.
- This phased method reduces financial risk and gives reliable data for future decisions.

Cross references

methodofenergysavingmeasure.pdf

itc-sustainability-report-2025.pdf

Supporting a healthier and greener India with our AI

Annual_Inventory2023-24.pdf

CircularNet: How Recykal built Asia’s largest circular economy marketplace using Google AI

APPENDIX

AI-Based Waste Reduction for Carbon Footprint Reduction: Tables, Technical Detail and

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A1. How waste creates carbon emissions

Supports Report section 1.

Pathway	**creates**	**Protocol**	**AI lever**
Embodied carbon	Material that is produced then wasted carries its production emissions (steel, plastic, cement, packaging, food).	Mostly Scope 3 (purchased goods) and Scope one-half for own production	Prevention, yield optimization, defect prediction
Landfill methane	Organic waste decays without oxygen and releases methane, about twenty-eight times as warm as CO2 over 100 years (Myhre et al., 2013).	Scope 3 (waste generated in operations) or Scope 1 for owned landfills	Segregation, composting and biogas routing
Rework and reprocessing	Rejected products need extra energy, materials, and machine time.	Scope 1 and 2	Early defect detection, predictive maintenance
Disposal logistics	Transport, incineration and treatment of waste emit CO2 and other gases.	Scope 1 and 3	Route optimization, local co-processing

A2. The five steps, goals, AI methods and headline KPIs

Supports Report section 2.

Step	Goal	Main AI methods	Headline
1. Baseline	Waste in tonnes and tCO2e per stream and per unit of output	Data validation, entity resolution, OCR and language models for documents, knowledge graph	Share of waste mass with measured quantity and emission factor
2. Prevent	Stop wasting before it exists	Supervised, unsupervised, deep and reinforcement learning, optimizations, graph search	Waste intensity (waste per unit produced)
3. Recover	Segregate and recycle unavoidable waste	CNN and instance segmentation, GAN augmentation, routing optimizations	Verified recycling rate and contamination rate
4. Convert	Make waste an input elsewhere	Knowledge graph, matching and transportation optimizations, A* and greedy search	Tonnes converted and tCO2e avoided per tonne
5. Verify	Prove and scale	Statistical tests, impact verification agent, audit logs	Verified tCO2e avoided and net financial benefit

A3. Which AI technique is used were

Lists every method named in the report: supervised, unsupervised, deep learning, reinforcement learning, search, optimization, knowledge graph, agentic AI and statistics.

Family	**Technique**	**Role in waste and carbon**
Supervised	Linear and regularized regression, Random Forest, XGBoost	Forecast waste, scrap, and energy per batch; estimate defect risk
Unsupervised	Isolation Forest, DBSCAN, K-Means, PCA, autoencoders	Find abnormal batches and operating regimes without labels
Deep learning	CNN, ResNet, YOLO, Mask R-CNN, U-Net	Defect inspection and waste sorting from images
Deep learning	RNN, LSTM, Transformer	Time-series forecasting of waste, demand and sensor drift
Deep learning	GAN and other generative models	Synthetic defect and waste images for rare classes; data augmentation
Reinforcement learning	DQN, PPO, safe and offline RL	Adaptive control of process settings and scheduling, tested in simulation first
Search	BFS, DFS	Trace anomalies and contaminated lots through the knowledge graph
Search	A*, Dijkstra, Greedy best-first	Lowest-cost paths for collection and recovery routing; fast candidate screening
Optimizations	Linear and mixed-integer programming, cutting-stock, vehicle routing, transportation problem	Yield, scheduling, collection routes, waste-to-user matching
Knowledge graph	Property graph, ontology, graph queries	Connect machines, batches, suppliers, waste events, and recovery sites
Agentic AI	Orchestrator plus specialized tool-using agents	Retrieve evidence, call models, compare actions, route approvals
Statistics	Control charts, hypothesis tests, FDR control, conformal prediction	Limit false alarms and missed events; verify savings

A4. Deep learning models and open-source resources

CNN, YOLO, Mask R-CNN, U-Net, RNN, LSTM, Transformer and GAN, with where each fits and its main caution.

Model	**Best use**	**Open-source or reference resource**	**Caution**
CNN / ResNet	Surface defect classification, material classification	ResNet (He et al., 2016); LeCun et al. (1998, 2015); PyTorch and TensorFlow reference implementations	Needs labelled images; performance drops with new lighting or camera angles
YOLO	Fast real-time defect and waste detection on lines and conveyors	YOLO family (Redmon et al., 2016); several open implementations [verify current maintained version]	Small objects and overlapping items are harder
Mask R-CNN	Instance segmentation, separating each waste item	Mask R-CNN (He et al., 2017); used on the TACO dataset (Proenca & Simoes, 2020)	Heavier compute than YOLO
U-Net	Pixel-level defect or contamination maps	U-Net (Ronneberger et al., 2015)	Requires pixel labels
RNN	Short sequences of sensor readings	Standard in deep learning libraries	Struggles with long dependencies
LSTM	Forecasting waste, demand and sensor drift; sequence anomaly detection	LSTM (Hochreiter & Schmidhuber, 1997); LSTM encoder-decoder anomaly detection (Malhotra et al., 2016)	Needs enough history; compare with gradient boosting and simple forecasts
Transformer	Long multivariate time series and document understanding	Attention mechanism (Vaswani et al., 2017)	Data hungry; can be overkill for small datasets
GAN	Synthetic images of rare defects and under-represented waste types	Generative adversarial networks (Goodfellow et al., 2014)	Synthetic data can hide bias; always test on real images

A5. Search algorithms: BFS, DFS, A*, Dijkstra, greedy and indexed search

How each works, where it is used, and its limitation.

Algorithm	works	framework	Limitation
BFS	Explores level by level from the anomaly node	Find directly connected machine, batch, material lot and inspections within a few hops	Large fan-out can return too much; limit depth
DFS	Follows one branch deeply before backtracking	Trace one suspect material lot through many processing stages to find affected batches	May go down irrelevant long branches; add depth and time limits
A*	Uses path cost so far plus a heuristic estimate of remaining cost (Hart et al., 1968)	Lowest-cost path in a dependency network or road network when the heuristic is admissible	Needs a good cost model and heuristic
Dijkstra	Finds least-cost paths without a heuristic (Dijkstra, 1959)	Baseline for routing when no good heuristic exists	Explores more nodes than A*
Greedy best-first	Always expands the node that looks closest to the goal	Fast approximate screening of promising recovery destinations or leads	No guarantee of lowest total cost; can pick a locally attractive but poor option
Indexed database search	Direct lookup by machine, batch, order or time window	Retrieve known records quickly, before using graph traversal	Does not reveal unknown relationships

A6. Agentic AI: agents, tools and autonomy levels

Which tasks the AI coordinator may do alone, and which need human approval.

Agent / module	**Task**	**Tools it may call**	**Autonomy level**
Waste Monitoring	Watch incoming waste, production and alert streams; open investigations	Read-only data queries, anomaly scores, dashboard	Autonomous (read-only)
Anomaly Investigation	Retrieve records, query the knowledge graph, run BFS or DFS, summarize evidence	Indexed database, graph query, maintenance and quality systems (read-only)	Autonomous (read-only)
Prediction	Request waste forecasts and defect-risk scores	Validated model endpoints	Autonomous (read-only)
Resource Optimizations	Compare alternatives for allocation, scheduling, collection and matching	Optimizations solvers, A* and greedy routines, simulators	Recommends only
Action Coordination	Assemble evidence, check approval rules, route to people or approved workflows	Ticketing, notification, workflow tools	Low-risk actions automated; others need approval
Impact Verification	Compare outcomes with baseline and record verified results	Statistical tools, reporting	Autonomous (read and report)

A7. Type I and Type II errors by decision

Shows which error is the more critical one for each kind of decision.

Decision	**Type I (false positive)**	**Type II (false negative)**	**More critical**
Automatic line stop or set-point change	Unnecessary stoppage, lost output, extra waste on restart	Waste continues until noticed	Type I (action is costly)
Alert to a human to investigate	Wasted inspection time, alert fatigue	Real anomaly missed	Usually Type II (cheap to check)
Reporting a waste or carbon saving	Claiming savings that did not happen (greenwashing, audit failure)	Real savings go unreported	Type I
Declaring waste "recycled"	Reporting recovery that did not occur	Under-reporting recovery	Type I
Blaming a supplier or material lot	Penalizing a good supplier, damaged relationship	Bad lot stays in use	Type I, but confirm before action
Hazardous waste screening	Extra testing and handling cost	Hazardous material mishandled: safety, legal and environmental harm	Type II
Product defect release check	Good product scrapped (waste and carbon)	Defective product ships	Depends on safety; Type II if safety-related

A8. Effect of alert threshold on errors and cost (illustrative)

Illustrative numbers: 10,000 batches a month, 200 truly abnormal. Costs are planning assumptions, not measured data.

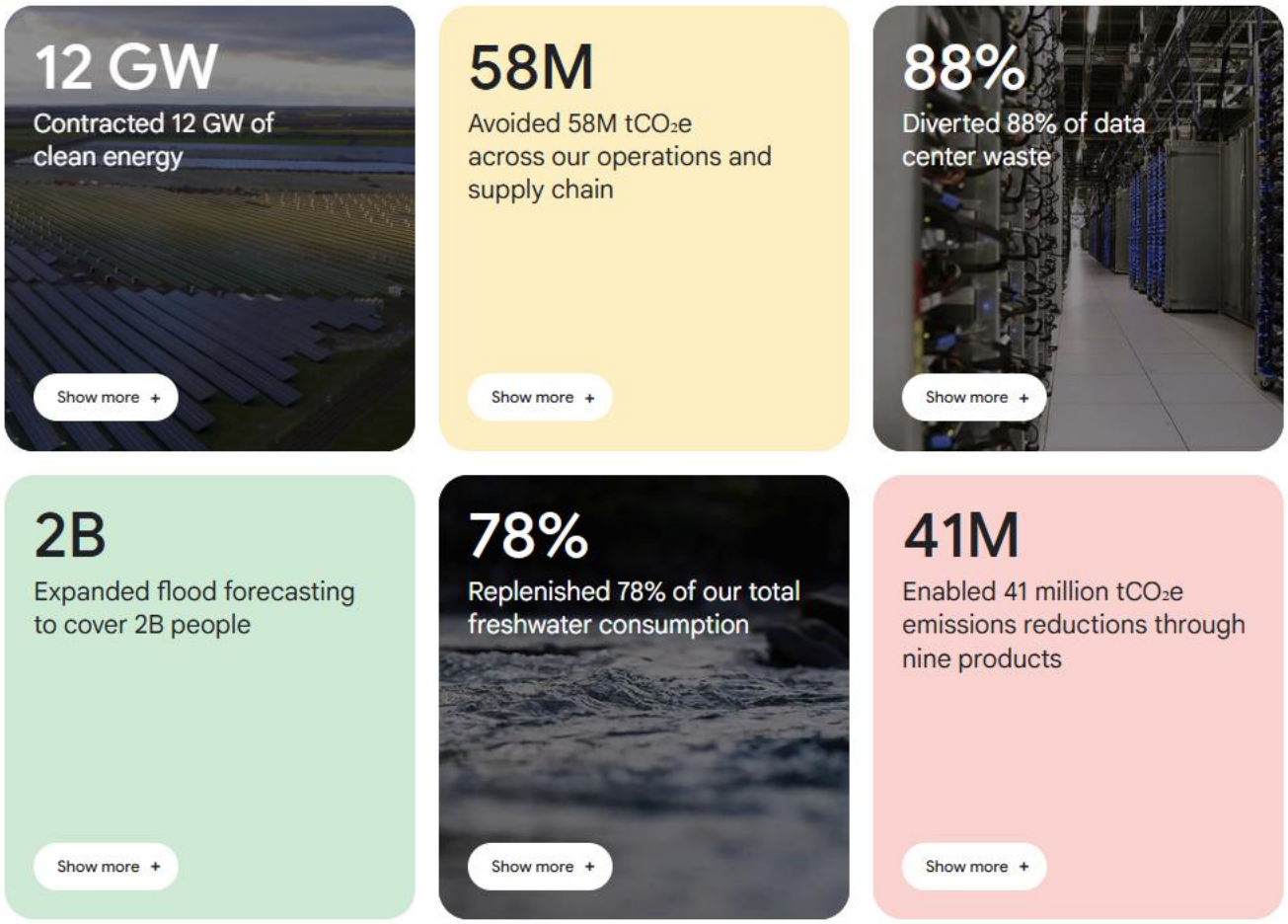
Threshold	**Caught**	**Missed (Type II)**	**False alarms (Type I)**	**Precision**	**Cost if alarm = Rs 5,000 and miss = Rs 50,000**	**Cost if alarm = Rs 3 lakh (auto line stop) and miss = Rs 50,000**
A: sensitive	180	20	490	27%	Rs 34.5 lakh	Rs 1,480 lakh
B: balanced	150	50	98	60%	Rs 29.9 lakh	Rs 319 lakh
C: strict	120	80	20	86%	Rs 41.0 lakh	Rs 100 lakh

A9. Case evidence from Google, Reliance, ITC, Adani, and others

All figures are self-reported by the organizations and are not guaranteed outcomes for another company.

Organization	Published result	Lesson for the conglomerate
Google (2025 Environmental Report)	Data center energy emissions down 12% in 2024 while electricity demand rose 27%; total emissions up 11%; over 8 GW of clean energy contracted; five products enabled an estimated 26 million tCO2e of reductions (Google, 2025a, 2025b).	AI efficiency can decouple output from emissions, but supply-chain growth can still raise total emissions.
Google / DeepMind (2016)	Up to 40% less cooling energy; 15% lower PUE overhead (Evans & Gao, 2016).	Validated prediction plus narrow, monitored actions on a live site.
Google CircularNet with Saahas Zero Waste	Open-source computer vision for plastic-waste characterization used in a sorting pilot (Google, 2024 [verify year]).	Start with one facility and measure recovery outcomes.
Reliance Industries	Non-hazardous waste diverted 883,130 t in FY 2025-26 (641,000 t prior year); chemically recycled plastic-based circular polymers; Scope 1 and 2 near 45.5 Mt CO2e (Reliance Industries, 2025, 2026).	Waste diversion grows fast, but energy dominates the footprint, so combine with energy decarbonization.
ITC	76,000 t plastic waste managed in FY 2024-25; 18 years solid-waste-recycling positive; WOW supports over 17,900 waste collectors (CSR Universe, 2025).	Collection networks and local partnerships are as important as technology.
Adani Group	8.52 Mt fly ash, 3.98 Mt slag, 1.3 Mt synthetic gypsum used as clinker replacement in FY 2022-23; Geoclinal co-processing; 12 ports with ZWL certification (Adani Group, 2023; IANS, 2024).	Industrial symbiosis gives the largest carbon lever in heavy industry; mind logistics.
Adani Dahanu	99.9% diversion claim with ash held in a pond for future use (Adani Energy Solutions, 2024).	Define "recycled" strictly and disclose it.
IKEA with Winnow	54% reported reduction in food waste globally (Winnow, n.d. [verify]).	Measurement itself changes behavior; do not assume the same result elsewhere.

Google 2026 Environmental report



10. Challenges, causes and mitigations

Supports Report section 5.

Challenge	**Why it happens**	**Mitigation**	**Owner**
Poor data quality and fragmented systems	Production, ERP, quality and waste records use different formats; sensors drift; weights are estimated	Data validation layer, entity resolution, weighbridges at priority streams, one pilot site, agreed data ownership	Data and plant IT
False alarms and alert fatigue (Type I)	Sensitive thresholds, thousands of signals, no context normalization	Calibration, cascades, persistence rules, FDR control, shadow mode, track precision and override rate (Section 10)	Data science and operations

Missed events (Type II), especially hazardous	Rare failures, drift, narrow training data	Cost-sensitive training, ensembles, confirmatory tests, audits of "no alert" samples	Safety and quality
Waste is a modest share of total emissions in heavy industry	Process energy and fuels dominate (Reliance reports about 45.5 Mt Scope 1 and 2)	Position waste as one pillar beside energy optimizations; target Scope 3 where waste matters more	Sustainability head
Scope 3 and supplier emissions	Gains inside the boundary hidden by supply-chain growth (Google total emissions up 11%)	Supplier batch data in the graph, supplier engagement, product and packaging redesign	Procurement
Weak downstream recovery infrastructure	No recyclers or users nearby; contaminated material	Partnerships with recyclers and local bodies, packaging redesign, regional matching	Facilities
Double counting and greenwashing risk	Loose definitions, shifting boundaries, predictions reported as savings	Strict definitions, fixed boundary, independent assurance, audit logs	Internal audit
AI platform footprint and cost	Compute and hardware use energy and money; Google data center electricity demand grew 27% in 2024	Use lighter models where adequate, run efficiently, count platform emissions in net benefit	IT and finance
Legacy and industrial control integration	Old PLCs and proprietary systems; cybersecurity exposure	Read-only integration first, network segmentation, security review, no agent access to control systems	OT security
Skills and adoption	Operators distrust black-box alerts; extra work	Co-design with plant teams, explanations with each alert, training, feedback loop for labels	Plant managers
Agent risks (hallucination, prompt injection, tool misuse)	Language models can produce unsupported claims or follow malicious text	Grounded citations, allow-listed tools, least privilege, evaluation and shadow mode (Section 9)	AI governance
Regulatory and compliance change	Waste, hazardous-waste and extended-producer-responsibility rules evolve	Compliance team reviews rules each quarter [verify current Indian rules]; human sign-off on compliance decisions	Compliance
Model drift	New materials, machines, seasons and suppliers	Drift monitoring, scheduled retraining, challenger models	Data science
Rebound and burden shifting	Savings encourage more output or move waste upstream	Normalized by output, review full chain, set absolute targets as well as intensity targets	Sustainability head
Unclear payback	Costs arrive early, benefits follow verification	Report investment and payback from the pilot itself, not from other companies' figures	Finance

A11. Expected outcomes by step and time

Planning ranges, not guarantees. Each must be confirmed by verification before it is reported as realized.

Step	Expected outcome	Indicator and target (planning)	When
1. Baseline	Trusted waste-carbon ledger; ranked list of top streams	Over 90% of pilot waste mass measured; emission factor and source for every stream	Months 1-3
2. Prevent	Lower scrap, fewer defects, less overproduction and expiry	5-10% waste-intensity reduction at the pilot site (planning target, subject to baseline)	Months 6-12
3. Recover	Higher verified recycling rate, lower disposal cost	Verified recovered tonnes with documents; contamination rate falling	Months 9-15
4. Convert	By-product use, lower landfill share, avoided virgin-material emissions	Tonnes converted per stream; tCO2e avoided per tonne with a stated substitution ratio	Months 12-18
5. Verify	Assured numbers and a scale-up decision	Verified tCO2e avoided and net benefit after platform costs, with confidence intervals	Months 10 and 18

A12. Roadmap and decision gates

Supports Report section 6.

Period	Key work	Decision gate
Months 1-3	Pick pilot site and streams; set boundaries; build waste-carbon baseline; integrate data; start knowledge graph	Baseline signed off by finance, operations, and sustainability
Months 3-9	Train forecasting, anomaly and defect models; run in shadow mode; tune thresholds with cost estimates; first optimizations use cases	Precision and recall meet agreed targets; false-alarm rate acceptable
Months 6-12	Vision-based segregation at one facility; recovery partnerships; routing; low-risk agent automation (reports, tickets)	Verified recovery with documents
Months 9-18	Waste-to-user matching and offtake agreements; quality blending; medium-risk workflows with approvals	Substitution ratios and transport emissions confirmed
Months 10 and 18	Formal verification with control group; assurance; governance review	Scale to more sites only if verified benefit justifies it